

LATENTSPRING: PHYSICS-INFORMED MOLECULAR FLOW MATCHING FROM ATOMIC COMPOSITION

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ABSTRACT

Generating three-dimensional molecules from atomic composition requires organizing atoms before their chemical connectivity is known. We introduce LatentSpring, a composition-conditioned flow-matching method with a harmonic source defined by latent spatial trees. The source uses atomic identities and covalent radii to sample correlated starting coordinates; averaging over trees gives an explicitly normalized density. A paired physical-training step then learns from local force-guided targets and transfers the update relative to a matched replay model. On 64 new compositions spanning 17–64 atoms, the harmonic source improves raw chemical-graph validity by 6.91 percentage points over an isotropic Gaussian source. The gain also appears on an independently initialized EGNN backbone. Component controls establish benefits from geometry self-conditioning and the paired physical update, which lowers energy and force under both eSEN and independent GFN2-xTB evaluation. On molecular rotors, complete escorted-work weights recover specified local target distributions with 58.5% lower distribution error than energy-only weights. The final generator uses one network with 128 evaluations per sample, and all reported generation metrics use its unoptimized coordinates.

1 INTRODUCTION

An atomic composition specifies which atoms are available, but leaves their chemical connectivity and three-dimensional arrangement unresolved. We study generation from this input, together with total charge and spin multiplicity. The model outputs atomic coordinates, from which chemical connectivity can be evaluated. This setting calls for both a useful spatial starting distribution and a way to incorporate physical information into the learned geometry.

Flow matching learns a transport from a source distribution to molecular structures (Lipman et al., 2023). An isotropic Gaussian provides a simple source, but its common coordinate scale carries little information about atom-specific geometry. Graph-conditioned harmonic sources supply such organization when a chemical graph is available. Our central question is how to obtain this benefit when the input contains atomic composition alone.

LatentSpring builds a spatial source by sampling an auxiliary tree over the atoms. Gaussian edge displacements, scaled by atomic radii, determine correlated coordinates. Averaging over possible trees yields a normalized mixture of spring networks with a tractable matrix-tree density. The tree organizes the starting coordinates; the learned flow determines the final geometry and its inferred chemical connectivity. We combine this source with an equivariant coordinate network and geometry self-conditioning, which refines a prediction using its own provisional geometry.

We also transfer local physical feedback into the generator. Forces evaluated at generated training structures define nearby physical targets. A physical model and a matched replay model learn from the same reference data and training schedule, and their parameter difference provides the update to the base generator. This produces one network for inference. Known local reference densities additionally make complete escorted-work weights available, allowing direct tests of sampling-bias correction on molecular torsional targets.

The contribution is a composition-based harmonic-mixture source, its integration with coordinate flow matching and paired physical learning, and controlled evidence for the resulting design. We test source

structure, self-conditioning and physical-update direction separately, evaluate new compositions and an independent backbone, and score raw geometries under two energy models.

2 RELATED WORK

Molecular flows and source distributions. Graph-conditioned methods such as ConfGF and GeoDiff generate conformations for a supplied chemical graph (Shi et al., 2021; Xu et al., 2022). FlowMol3 supplies our equivariant coordinate network (Dunn & Koes, 2026): rotating its input rotates its predictions, and relabeling identical atoms relabels the outputs. We reuse its geometry self-conditioning modules. ET-Flow uses harmonic sources constructed from a supplied molecular graph (Hassan et al., 2024). More general source adaptation includes guidance under a frozen field (Wang et al., 2026) and learned conditional sources (Kim et al., 2026). LatentSpring constructs its molecular source from atomic composition by integrating over auxiliary spatial trees.

Distributions over trees. Factored tree distributions and tractable tree learning provide the probabilistic basis for our construction (Meila & Jaakkola, 2000; Duan & Dunson, 2023). Gaussian mixtures over random spanning-tree differences have also been used in imaging inverse problems (Everink, 2026). Here, isotropic edge vectors and atomic length scales define a density on centered molecular coordinates. Our Gaussian, shell and covariance controls test which aspects of this structure help the molecular flow.

Physical feedback and nonequilibrium work. OMol25 provides molecular structures, energies and forces across electronic conditions (Levine et al., 2025). Energy-weighted flow matching (Dern et al., 2025) and target-density training with importance sampling (Midgley et al., 2023) incorporate physical information into generative models. Physical-reward fine-tuning offers a further route for molecular diffusion models (Zhou et al., 2025), and physical simulations can supply fine-tuning data for protein conformation samplers (Lu et al., 2024). Jarzynski’s equality relates nonequilibrium work to equilibrium free-energy differences; Hummer–Szabo reconstruction applies it to single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport extends this relation to imposed flow fields with the corresponding volume correction (Vaikuntanathan & Jarzynski, 2008). Targeted perturbation incorporates coordinate maps into free-energy reweighting (Jarzynski, 2002), and learned escorted protocols improve multistate free-energy estimation (Holdijk et al., 2026). We use these work identities with normalized local references to construct physical training targets for composition-conditioned generation.

3 A HARMONIC SOURCE WITH LATENT CONNECTIVITY

Our model takes an atomic composition and its electronic state as input and generates three-dimensional atomic coordinates. The input records the element of each atom through its atomic number z_i , the total charge Q in elementary-charge units, and the spin multiplicity S . We collect these quantities as $c = ((z_i)_{i=1}^N, Q, S)$, retaining every atom, including repeated elements. A neutral singlet has $Q = 0$ and $S = 1$. For example, $\text{C}_2\text{H}_6\text{O}$ specifies two carbon atoms, six hydrogens and one oxygen, while allowing different molecular connectivities. Chemical bonds are inferred from generated coordinates for evaluation; they are not supplied as training labels.

For the N input atoms, their positions form a matrix $X \in \mathbb{R}^{N \times 3}$. Its row x_i contains atom i ’s three coordinates, measured in angstroms. We remove the molecule’s arbitrary absolute location by subtracting the arithmetic mean of its coordinates. The resulting centered coordinate space is $\mathcal{H}_N = \{X : \sum_i x_i = 0\}$; it has $3(N - 1)$ independent coordinates.

A random scaffold for the starting coordinates. We first connect the atoms by an auxiliary spanning tree T , a connected graph with $N - 1$ edges and no cycles. Its edges organize the starting cloud and do not specify the final chemical bonds. Each possible edge (i, j) receives a positive weight $a_{ij} = a_{ji}$, called its affinity, for the connection between atoms i and j . A tree’s probability is proportional to the product of its edge affinities:

$$p(T \mid c) = \frac{\prod_{(i,j) \in T} a_{ij}}{\tau(c)}, \quad \tau(c) = \sum_T \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

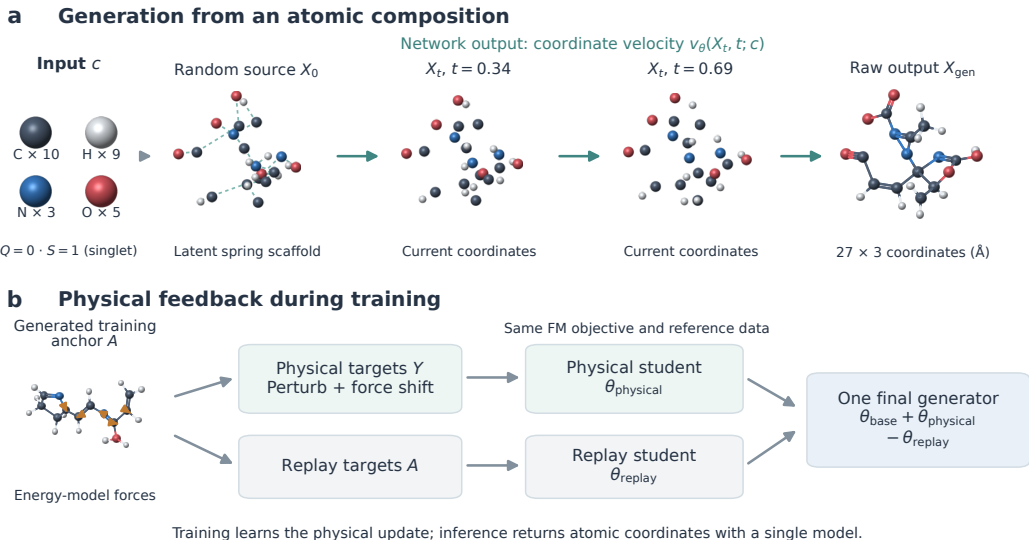


Figure 1: LatentSpring’s inputs, outputs and physical learning. (a) An actual recorded flow for $C_{10}H_9N_3O_5$. The source supplies random coordinates X_0 ; the network reads current coordinates, flow time and composition, and predicts movement vectors. Solid bonds are inferred only for the displayed final output; dashed source edges are auxiliary springs. (b) A generated training anchor from a separate composition supplies local physical targets and matched replay targets. Their learned parameter difference updates one generator. All displayed molecular coordinates are unoptimized.

Here $\tau(a)$ sums those products over all spanning trees, making their probabilities sum to one. We use factorized affinities $a_{ij} = u_i u_j$, with $u_i = 0.05$ for H and F/Cl/Br/I, and otherwise $u_i = \max(1, \text{default valence}(z_i) - 1)$. The scalar u_i controls how readily atom i appears in tree edges. These settings favor fewer connections for hydrogen and halogens, which usually occupy terminal positions. Default valence is the element’s tabulated bonding capacity, counting bond order; no molecule-specific bonds are supplied.

Gaussian edge laws and sampling. For each tree edge, we draw a random three-dimensional displacement between its atoms. The characteristic length ℓ_{ij} is the sum of their covalent radii, which estimates a typical single-bond separation. Set $\sigma_{ij}^2 = \ell_{ij}^2 \exp(s^2)/3$, with $s = 0.2$ fixed throughout. The displacement $d \in \mathbb{R}^3$ has probability density

$$h_{ij}(d) = (2\pi\sigma_{ij}^2)^{-3/2} \exp\left[-\frac{\|d\|^2}{2\sigma_{ij}^2}\right]. \quad (2)$$

Here $\|d\|$ is its Euclidean length and σ_{ij} is the standard deviation along each coordinate axis. All directions are equally likely; the radius-based scale sets the typical separation. This scale also matches the edge covariance of the lognormal shell control. We sample a tree with the weighted Wilson algorithm, draw its independent edge vectors, place atoms by adding these vectors along tree paths, and subtract the coordinate mean. A new tree is drawn for each sample. Rotating the molecule or relabeling identical atoms leaves this probability law unchanged.

The distribution after averaging over trees. The source density $q_0(X | c)$ describes the probability of starting coordinates X given the atomic input c . Averaging over the unobserved tree gives a mixture of harmonic networks:

Proposition 1 (Normalized source density). *For normalized isotropic edge densities,*

$$q_0(X | c) = N^{3/2} \frac{\tau(a_{ij} h_{ij}(x_i - x_j))}{\tau(a)} \quad (3)$$

integrates to one over the centered coordinate space $\mathcal{H}_N$.

In the numerator, each affinity is multiplied by the density of its observed edge displacement. The factor $N^{3/2}$ accounts for expressing independent edge vectors as centered atomic coordinates. The matrix-tree theorem evaluates each tree sum as a matrix determinant, avoiding enumeration of all trees. The normalizing factor is derived in Appendix G.

The term *harmonic* refers to the quadratic spring energy associated with each Gaussian edge law. These virtual springs define a spatially organized random starting distribution. The learned flow establishes the final molecular geometry.

4 INTEGRATION WITH MOLECULAR FLOW MATCHING

The flow converts random coordinates $X_0 \sim q_0(\cdot | c)$ into generated coordinates $X_{\text{gen}} \in \mathcal{H}_N$, conditional on the fixed atomic input c . At each step, its network reads the current coordinates X_t , the progress variable t , and c , and predicts a coordinate velocity $v_\theta(X_t, t; c) \in \mathbb{R}^{N \times 3}$: one movement vector for every atom. Integrating these predictions from $t = 0$ to $t = 1$ produces the molecular coordinates. The latent tree is used to sample X_0 ; it is not an additional chemical graph supplied to the network. Energy and force evaluations enter physical training below, while generation uses the resulting network alone.

Constructing the training input and target. We draw X_0 from the harmonic source and take reference coordinates X_1 from OMol25. Before pairing them, we match atoms of the same type and rotate the structures to shorten their coordinate displacement. We then apply a shared random rotation, uniform over orientations, and a shared random relabeling of identical atoms. This preserves the source distribution (Appendix H).

The interpolation coordinate $t \in [0, 1]$ describes progress from the source ($t = 0$) to the reference ($t = 1$), rather than physical simulation time. The training path is $X_t = (1 - t)X_0 + tX_1$. Its velocity is the constant matrix $V^* = X_1 - X_0$, giving a supervised target for each atom’s movement.

Predicting with a provisional geometry. The trainable weights are denoted by θ . The implementation retains FlowMol3’s coordinate-shaped output head $D_\theta(X_t, t; c)$ and trains its increment $D_\theta - X_t$ as the velocity. We subtract the mean of these movement vectors to preserve centered coordinates. Geometry self-conditioning gives the network its own provisional endpoint as an additional input. A first evaluation predicts velocity $v^{(0)}$, yielding the endpoint estimate $\hat{X}_1 = X_t + (1 - t)v^{(0)}$. The factor $1 - t$ is the remaining interpolation interval. A second evaluation uses this estimated geometry to predict a refined velocity $v^{(1)}$. Both evaluations share network weights.

We supervise both predictions so that the provisional geometry also learns from the reference displacement. Their mean squared velocity error is

$$\mathcal{L}_{\text{FM}} = \frac{1}{2} \mathbb{E} \left[\frac{\|v^{(0)} - V^*\|_F^2 + \|v^{(1)} - V^*\|_F^2}{3N} \right]. \quad (4)$$

The expectation $\mathbb{E}$ averages over training structures, source draws and random values of t . The squared Frobenius norm $\|\cdot\|_F^2$ sums errors over all $3N$ coordinate components; dividing by $3N$ gives equal weight to each molecule. The factor $1/2$ averages the two predictions. Gradients pass through both evaluations. Atomic categories and charge inputs remain fixed, and edge inputs carry a constant no-bond state. The FlowMol3 source comparisons share this coordinate-only objective and the same two-pass self-conditioning architecture.

Generation repeatedly follows the learned velocity toward $t = 1$. We use 32 midpoint integration steps, each with two velocity evaluations. The first uses the current coordinates and the interval’s midpoint time; the second uses the resulting half-step coordinates at that same time. Two self-conditioning passes per velocity evaluation give $32 \times 2 \times 2 = 128$ primitive network evaluations per attempted sample, followed by $0.025 \text{ \AA}$ centered Gaussian noise. The FlowMol3 source comparisons use this same sampler. We evaluate its raw coordinates directly, without subsequent energy selection or geometry optimization.

4.1 PHYSICAL FEEDBACK THROUGH PAIRED TRAINING TARGETS

Physical learning uses energy and force information during training to improve the coordinates produced by the final network. A frozen harmonic generator first produces training structures, called anchors and denoted by A , on compositions excluded from evaluation. For each anchor that passes the chemical-graph checks, we draw nearby coordinates from a small centered Gaussian cloud $q_A = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. Here σ_A sets the size of local perturbations and I is the identity covariance on the centered space.

The teacher uses the learned eSEN energy model, with energy E_+ and force F_+ , averaged over mirror-related coordinates to enforce reflection symmetry (Appendix K). Locally linearizing this energy shows that Boltzmann weighting shifts the Gaussian’s mean by $\delta_A = \sigma_A^2 F_+(A)/(k_B T_{\text{phys}})$, where k_B is Boltzmann’s constant and $T_{\text{phys}} = 300$ K is the teacher temperature. Applying this displacement to each nearby structure gives $Y = X + \delta_A$, called a force escort. We choose the perturbation scale to bound the displacement, and fix both quantities before drawing candidates. Appendix K derives this local Boltzmann approximation and specifies its geometric restriction.

The already-trained harmonic generator has weights θ_{base} . We make two copies of these weights. The physical copy learns from the force-shifted targets and produces θ_{physical} . The replay copy follows the same training schedule using the original generated anchors, producing θ_{replay} . Both copies also receive the same updates on original OMol25 references. Replay measures the change caused by refitting generated structures without their physical displacement. We construct the final network weights by adding the difference between these continuations to the original model:

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}}). \quad (5)$$

The subtraction defines a physical-versus-replay update direction. Its addition to θ_{base} is parameter arithmetic (Ilharco et al., 2023), applied elementwise to matching weight tensors with coefficient one. The result is one neural generator; the two training copies are not evaluated during generation. Direct and paired-update results are reported in Appendix K.

Correcting the distribution of local candidates. Complete escorted-work weighting (Vaikuntanathan & Jarzynski, 2008) provides a local extension that favors lower-energy structures while retaining the Gaussian restraint around each anchor. Because the escort changes where candidates are sampled, energy differences alone do not give the target weights. For candidates retaining the anchor’s inferred connectivity, define

$$W_A = E_+(Y) - E_+(A) + k_B T_{\text{phys}} [\log q_A(X) - \log q_A(Y)]. \quad (6)$$

The work W_A has energy units. Its first term measures the energy change; the logarithmic density ratio corrects the sampling shift. The map is a fixed translation, so its volume-change factor is one. Weights $\exp[-W_A/(k_B T_{\text{phys}})]$ estimate the local target’s normalizing constant and determine how often each candidate is used for training. We compare uniform and complete-work training weights, and use the less expensive force update as the primary candidate for the new-composition physical evaluation. Appendix K gives the target definition, particle construction and training controls.

5 EXPERIMENTAL DESIGN

Task and training. We study adaptation of a pretrained molecular flow to coordinate generation from neutral singlet organic compositions. Each composition contains C and H and uses elements from H/B/C/N/O/F/Si/P/S/Cl/Br/I. Source-model training covers 8–40 atoms; evaluation extends to 64 atoms. The implementation supports up to 200. Two continuations start from a shared pretrained checkpoint. Each uses 3,000 distinct OMol25 structures and 3,000 optimizer updates, with identical data and initialization across its source variants. The vector field has 5,904,369 parameters. The backbone learning rate is $2 \cdot 10^{-5}$ and the self-conditioning modules use $3 \cdot 10^{-4}$.

Training structures pass the graph assay and a geometric filter requiring a bounded-edge, degree-capped scaffold. This filter is specified in the data-selection protocol. The models receive the selected coordinates; the scaffold is used for selection only. All source variants use the same filtered subset.

Composition panels. The source studies use 12 development compositions, ten additional compositions, 24 fresh compositions and a broader 64-composition panel. The latter three are absent

from both processed OMol25 corpora by exact atomic-count checks. Reference coordinates qualify the evaluation conditions and are excluded from fitting. The pretrained studies reuse the same two fitted continuations; the independent EGNN study trains two new initializations. Selection details and source controls are given in Appendix C.

Metrics and uncertainty. The first two panels use 64 samples per composition and the 24-composition test uses 32. Geometric validity requires a connected contact graph and no short-distance overlap, with thresholds of 1.25 and 0.6 times the summed covalent radii. Graph support additionally requires RDKit to construct a connected, sanitized graph with the specified total charge. We report support per attempted output, including failed samples and validator exceptions in the denominator. Distinct perceived connectivities within each composition measure graph diversity; timings record the full generation procedure. Appendix B gives the selection and perception settings.

Unless stated otherwise, 95% confidence intervals use paired bootstrap resampling of generation draws, holding the fitted models and compositions fixed. We also bootstrap compositions to describe variation across the evaluation panel. Physical scores are computed at the saved raw coordinates using the declared electronic conditions and approximate potentials.

6 RESULTS

6.1 EFFECT OF THE LATENT SPRING SOURCE

With backbone, initialization and training data matched, harmonic sources improve chemical-graph validity over isotropic Gaussian sources by 5.86 percentage points on 12 development compositions, 8.83 points on ten additional compositions, and 8.20 points on the 24-composition panel below. A Gaussian with approximately matched covariance also performs worse, supporting a contribution from mixture structure beyond the tested second-moment approximation. Appendix C reports the detailed source, shell and covariance controls.

6.2 GENERALIZATION ACROSS MOLECULAR SIZES

We evaluate the frozen source models on 64 further compositions, with 16 in each of four size ranges from 17 to 64 atoms. All are absent from both processed OMol25 corpora and from the earlier generation panels. The models receive no additional training. Metadata ranks select each composition after the same reference-validity checks, with 32 draws per composition and model continuation.

Across all 64 compositions, Gaussian sources yield 24.85% graph validity and harmonic sources yield 31.76%. Appendix D gives the full size-stratified rates.

The harmonic source improves the pooled rate in every size range. Overall it adds 6.91 percentage points, with a size-stratified composition-bootstrap 95% interval of [4.74,9.03]. The two continuations improve by 11.18 and 2.64 points, respectively. Absolute validity decreases for larger molecules, while the source advantage remains positive in the aggregate. These results extend the source comparison beyond the original 17–28-atom confirmation range.

6.3 INDEPENDENT BACKBONE AND GENERATOR OBJECTIVES

We repeat the source comparison on an independent EGNN backbone and compare with position-conditional EDM and GAGA (Hooeboom et al., 2022; Qu et al., 2026). All four methods train from scratch on the same 20,000 OMol25 rows, with identical architecture and initial weights within each of two repetitions. Each receives 30,000 updates at batch size 32. The FM arms share their training path and atom pairing; their source distribution is the only difference. Appendix F gives the protocol and the transferred GAGA truncation setting.

The harmonic source increases FM validity by 1.34 percentage points, with a composition-bootstrap 95% interval of [0.12,2.52] for the signed difference. This comparison tests source transfer independently of pretrained backbone history. The diffusion objectives reach higher validity in this setting. The matched FM contrast isolates the source contribution; comparisons with diffusion also change the probability path, objective and sampler.

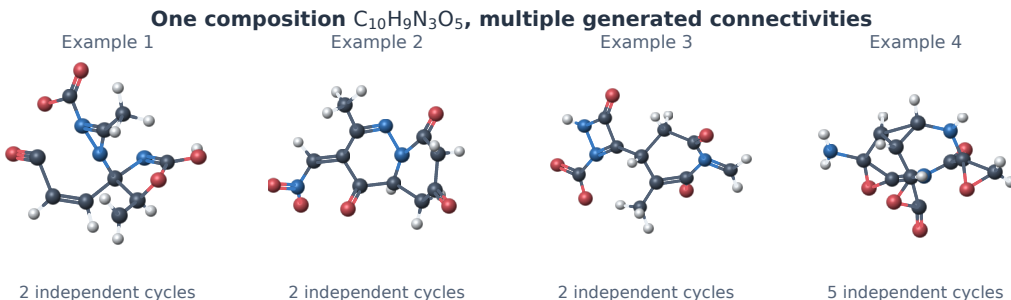


Figure 2: Raw outputs of the primary force-updated LatentSpring model with the same atomic composition and different inferred connectivities. The source trees are acyclic, while the generated structures can contain cycles. Examples are the first four valid, distinct-connectivity draws for the first CHON composition with valid outputs in continuation 1 of the 24-composition panel. No energy ranking or geometry optimization selects or changes them. The number of independent cycles is the inferred edge count minus atom count plus one.

Table 1: Chemical-graph validity on the same 64 compositions, with 2,048 attempts per initialization and 128 network evaluations per output. All models share the 2.38-million-parameter EGNN architecture and have no pretraining.

Method	Valid, seed 1	Valid, seed 2	Pooled rate (%)
Gaussian FM	182	218	9.77
Harmonic FM	223	232	11.11
EDM	301	216	12.62
GAGA	321	249	13.92

6.4 NEW COMPOSITIONS AND INDEPENDENT PHYSICAL EVALUATION

We evaluate frozen Gaussian, harmonic and physical-update models on 24 new compositions, selected by fixed metadata ranks after reference qualification. The force-only update is chosen before this evaluation. Eight compositions fall in each of the 17–20, 21–24 and 25–28 atom ranges, with zero exact composition overlap in either processed corpus. Each model generates 32 samples per composition and continuation, giving 6,144 raw outputs across four models. Selection details are provided in Appendix B.

The harmonic source improves graph validity by 8.20 percentage points over Gaussian (95% CI [5.08, 11.33]), with gains in both continuations. The force update changes graph validity by +0.72 points relative to harmonic, with interval [−0.98, 2.34], consistent with a similar support rate. Every accepted output has distinct perceived connectivity within its composition in these draws.

Energy and force scores. Every output is evaluated at fixed coordinates with eSEN and gas-phase GFN2-xTB single-point energies and gradients (Bannwarth et al., 2019). GFN2 provides an independent potential, used only for evaluation. Four of the 6,144 generated-geometry GFN2 calculations fail; all four concern graph-invalid outputs. Calculations succeed for every graph-valid output, and reference inversion checks pass. Including reference checks, evaluation costs 12,384 eSEN queries and 6,240 GFN2 attempts. eSEN scores use the parity averages defined in Section 4. Force RMS is the square root of the mean squared atom-force norm.

The primary contrast uses paired outputs that pass the graph assay for both models, with successful electronic calculations. Such pairs are available for all 24 compositions in both continuations. Energy and force improve under both potentials. The same direction holds when including all outputs for eSEN and all jointly successful calculations for GFN2. For the source-only harmonic–Gaussian comparison, aggregate energy also decreases, with a larger effect in one continuation than the other.

Table 2: Fresh 24-composition graph validity, 768 attempts per continuation. All rates describe raw outputs before any structural relaxation. All methods use 128 primitive network evaluations per attempted output.

Model	Continuation 1	Continuation 2	Pooled rate
Gaussian reference	332	352	44.53%
LatentSpring harmonic	435	375	52.73%
With force update	439	382	53.45%
With complete-work update	426	383	52.67%

Table 3: Primary force-update change versus the frozen harmonic model on paired jointly valid outputs. Both continuations improve under both potentials. Brackets give paired 95% confidence intervals.

Potential	Energy change (eV/atom)	Force-RMS change (eV/Å)
eSEN	-0.01308 [-0.01805, -0.00817]	-0.34793 [-0.41498, -0.28478]
GFN2	-0.01168 [-0.01616, -0.00724]	-0.35620 [-0.42983, -0.28714]

The complete-work and force-only updates perform similarly within sampling uncertainty. Their energy difference among jointly valid outputs is -0.00413 eV/atom under eSEN, with interval $[-0.00903, 0.00078]$, and -0.00363 under GFN2, with interval $[-0.00792, 0.00075]$. Force and graph differences also have intervals spanning zero. The force-only update therefore offers comparable physical scores with a lower teacher-preparation cost.

6.5 EVIDENCE FOR THE TRAINING COMPONENTS

Self-conditioning improves graph validity by 7.23 percentage points at equal optimizer updates (95% interval $[4.69, 9.77]$) and by 12.43 points at equal training-pass count ($[9.70, 15.23]$). Both continuations improve under both controls. The paired physical update lowers energy by 0.03232 eV/atom relative to a direct update with the same parameter norm. Removing the force displacement from otherwise matched local targets raises energy by 0.01939 eV/atom. These energy contrasts use paired outputs valid under both methods, with the same signs in the all-output comparisons. Appendix N reports the complete budgets, validity rates, force scores and confidence intervals.

6.6 COMPLETE-WORK CORRECTION ON MOLECULAR ROTORS

To test the work correction directly, we select three methyl rotors from training molecules in fixed data order, using geometry alone. Each rotor turns the three hydrogens of a methyl group around its bond axis while holding the other internal coordinates fixed. The rotation angle ϕ therefore specifies a one-dimensional family of three-dimensional molecular structures. A periodic interpolant of 512 GFN2 energies supplies their potential $U(\phi)$; $U(0)$ is the reference energy at angle zero. The target is $\rho(\phi) \propto q_0(\phi) \exp[-(U(\phi) - U(0))/kT]$, where q_0 is a normalized von Mises reference of concentration one: a smooth circular distribution favoring angles near zero. The product retains this reference restraint and gives greater weight to lower-energy angles. Here $kT = k_B T_{\text{phys}}$ at 300 K.

Three fixed angle transformations, including an unchanged-angle control, distort the starting distribution. Complete work corrects both the reference-density change and the local expansion or compression of angular intervals. We compare these weights with energy-only weights, weights omitting the volume correction, and unweighted samples. Each setting uses 128 repetitions at sample counts from 8 to 2,048; Appendix L specifies the maps and work formula.

Total variation measures disagreement between the estimated and target angular probabilities, with zero indicating agreement. Free-energy RMSE measures error in the logarithm of the target normalizer, expressed in energy units.

At 2,048 samples, complete work has lower mean total-variation error than both incomplete-weight controls in all six nonidentity cases. Across these cases, mean errors are 0.0636 for complete work, 0.1532 for energy-only weights and 0.1134 without the Jacobian, a 58.5% reduction relative to energy-

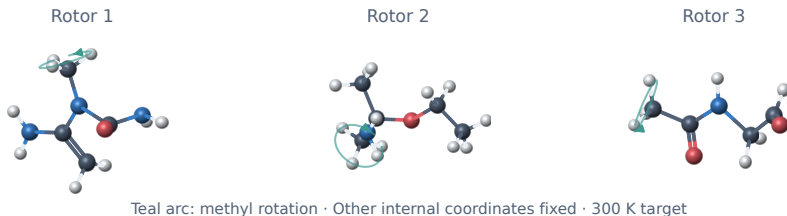
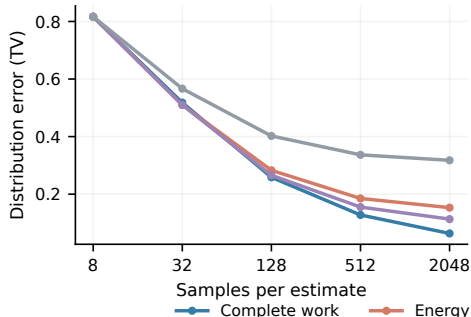
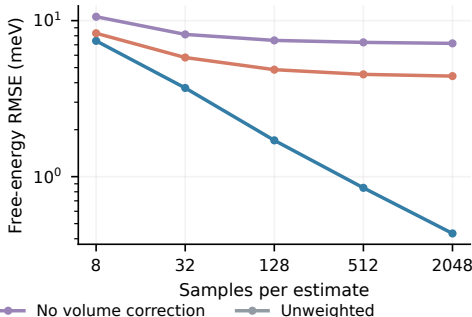
a Controlled methyl rotation on three molecules**b Recovering angular probabilities****c Estimating the normalizer**

Figure 3: Molecular work correction. (a) The three reference structures and their prescribed methyl rotations. (b,c) Distribution and normalizer errors, averaging all three molecules and both nonidentity escorts, with 128 repetitions per case. Complete work removes the sampling bias left by energy-only and incomplete weights. Unweighted endpoints have no free-energy estimator. Each target is a restrained molecular rotor at 300 K.

only weighting. Complete-work free-energy RMSE ranges from 0.158 to 0.734 meV. Independent density-ratio quadrature also recovers the specified target normalizer in all nine molecule/escort cases. The identity-escort control makes the three weighted methods coincide. Together, the distribution and normalizer results show the role of complete work accounting on these molecular energy curves.

7 DISCUSSION

LatentSpring organizes the starting distribution through latent connectivity and transfers local physical information into a single molecular flow. Controlled comparisons support the source, geometry feedback and physical-update choices. The evidence covers neutral organic compositions, two pretrained-model continuations and two independent EGNN initializations; intervals condition on these fitted models. Chemical-graph validity measures structure, energy and force readouts use approximate potentials, and the work experiments address specified local ensembles. Broader chemistry and recovery of full molecular equilibrium remain open directions.

REPRODUCIBILITY STATEMENT

The project includes code, model configurations and data-selection manifests. These specify random seeds, checkpoint hashes, evaluation settings and every attempted output. We report training, physical preparation and generation costs separately, and provide tests for source normalization, symmetry, density gradients and work identities. Source data and pretrained backbones retain their original licenses and access requirements.

AI USE STATEMENT

OpenAI Codex and Claude Code assisted literature review, hypothesis and experiment design, mathematical derivation and proof writing, code implementation, data preprocessing, testing, experimental analysis, and manuscript drafting and editing.

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A READER’S GUIDE TO THE NOTATION

The method has three stages: construct random starting coordinates from the atomic input, learn a flow toward molecular reference structures, and use physical targets to update the trained network. The symbols below refer to these stages.

What the model receives and returns. The requested input c fixes the atoms and electronic state. Randomness enters through the sampled starting coordinates X_0 . Each network call receives the current coordinates X_t , flow time t , and c ; the two self-conditioning passes internally construct a provisional geometry. The predicted output is a velocity matrix with one three-dimensional movement vector per atom. Numerical integration and the specified terminal noise produce the returned coordinate matrix X_{gen} . The original atomic identities, charge and spin accompany these coordinates. Chemical connectivity is inferred separately for evaluation.

Symbol	Meaning
N, z_i, Q, S, c	Number of atoms, element-identifying atomic numbers, net charge, and spin multiplicity; c collects the atomic input.
$X, x_i, \mathcal{H}_N$	Atomic coordinate matrix, the position of atom i , and the space where the mean atomic position is zero. Coordinates are in angstroms.
T, a_{ij}, u_i	Auxiliary spatial tree, its edge affinities, and the atom-dependent factors defining those affinities. These are source variables.
$\ell_{ij}, \sigma_{ij}, h_{ij}$	Characteristic separation from covalent radii, Gaussian displacement scale, and the probability density of one edge vector.
$\tau(a), q_0(X c)$	Sum of tree weights and the normalized distribution of starting coordinates after averaging over trees.
$X_0, X_1, X_t, X_{\text{gen}}, t$	Starting, reference, intermediate and returned coordinates; t labels generation progress, not physical time.
V^*, v_θ, D_θ	Target velocity, predicted velocity, and the coordinate-shaped network head whose centered increment $D_\theta - X_t$ is trained as the velocity.
$v^{(0)}, v^{(1)}, \hat{X}_1$	First and second velocity predictions, and the provisional endpoint passed between them during self-conditioning.
$\mathbb{E}, \ \cdot\ _F^2$	Average over the specified random draws, and the sum of squared entries of a coordinate or force matrix.
$\theta_{\text{base}}, \theta_{\text{physical}}, \theta_{\text{replay}}, \theta_{\text{final}}$	Original trained weights, physical and replay continuations, and the resulting weights used for generation.
$A, q_A, \sigma_A, \delta_A$	Generated training anchor, Gaussian neighborhood around it, neighborhood scale, and force-derived displacement.
E_+, F_+	Mirror-symmetrized teacher energy and force, in eV and eV/Å. The force is the negative coordinate gradient of this energy.
$k_B T_{\text{phys}}, W_A$	Thermal energy at the teacher temperature and generalized work used to weight local candidate structures.
Φ, J, Z	An imposed coordinate transformation, its local volume-change factor, and the integral that normalizes a target density.

Table 4: Symbols used in the source construction, flow model and physical update. The tree T and physical temperature T_{phys} denote different quantities.

Structural and physical readouts. Geometric validity checks that all atoms belong to one contact-connected structure and that no pair has an excessive short-distance overlap. Chemical-graph validity also requires a chemically consistent bond graph inferred from those coordinates. A pair of outputs is *jointly valid* when both pass these checks; their inferred chemical graphs can differ. By contrast, the local physical teacher’s same-graph restriction keeps the anchor’s labeled bond orders, formal charges and radical counts fixed.

Energy changes are compared within the same atomic composition and divided by its atom count. Force RMS is $[N^{-1} \sum_{i=1}^N \|F_i\|_2^2]^{1/2}$, the root mean squared magnitude of the atomic force vectors. Lower force RMS indicates smaller residual forces, while the energy score measures relative energetic preference under the chosen potential.

B DATA SELECTION AND STRUCTURAL EVALUATION

Training data are selected in a fixed shuffled order from the processed OMol25 training corpus, excluding the evaluation compositions. The first 3,000 eligible structures form each continuation’s training set. Eligibility requires the atom, element and electronic-state conditions in Section 5, the structural assay below, and a geometric scaffold filter.

Table 5: Chemical-graph validity per attempted output. All primary arms share the geometry-SC backbone and 128 primitive network evaluations. The additional panel uses frozen models selected after the development comparison.

Panel	Continuation	Gaussian	Shell tree	Harmonic tree
Development (12)	1	357/768	417/768	436/768
	2	377/768	342/768	388/768
Additional (10)	1	326/640	—	413/640
	2	349/640	—	375/640

The scaffold filter considers pairs whose distance divided by the sum of covalent radii lies strictly between 0.6501 and 1.1999. Kruskal’s algorithm constructs a minimum normalized-distance spanning tree from these pairs, greedily adding short edges that do not create a cycle. A structure is accepted if the tree is connected and its degrees are at most 1 for H/F/Cl/Br/I, 4 for B/C/N/Si, 3 for O, and 6 for P/S. The scaffold is used only to select training coordinates; the molecular flow uses its own sampled harmonic source.

Structural evaluation uses RDKit 2025.03.2, a toolkit for representing molecules and checking chemical consistency. After the contact-connectivity and overlap checks, `DetermineBonds` assigns bonds at the specified total charge, with `covFactor=1.25`, `useVdw=True`, `allowChargedFragments=True`, and `embedChiral=False`. Its sanitization step checks valence and related graph-consistency rules. Sanitization must succeed, the assigned formal charges must sum to the specified charge, and the perceived molecule must have one connected component. Graph diversity counts canonical SMILES, standardized text encodings of molecular connectivity, after removing stereochemical tags and explicit hydrogens. Evaluator errors and unsupported outputs remain in the attempted-sample denominator.

The 24-composition confirmation panel. We next evaluate the frozen Gaussian, harmonic and two physical-update models on 24 new compositions. The force-only update is chosen as the primary physical candidate before this evaluation. Starting from a pool of 256 distinct compositions outside the earlier candidate sets, reference qualification accepts 121. Fixed metadata ranks select eight compositions from each of the 17–20, 21–24 and 25–28 atom bins. All 24 are absent from the processed OMol25 training and validation corpora. Reference coordinates are used solely for qualification. Each model generates 32 samples per composition in each continuation, giving 6,144 raw outputs in total.

C SOURCE-DEVELOPMENT AND COVARIANCE COMPARISONS

Composition panels. Reference qualification accepts 31 of 78 metadata-eligible compositions. Of these, 21 had appeared in earlier development studies. We use a fixed 12-composition subset for the source comparison and all 10 remaining compositions for subsequent confirmation. The latter have 17–28 atoms and are absent, by exact composition count-vector checks, from 3,902,107 processed training rows and 39,415 validation rows. Reference geometries serve only to qualify the evaluation conditions. Both panels use the same two fitted continuations. A further 24-composition test is described in Section 6.4.

Harmonic trees improve graph validity over isotropic Gaussian sources by 5.86 percentage points on the development panel (95% CI [2.67,8.98]; composition interval [0.52,12.17]). Both continuations improve, with a larger gain in the first. The harmonic source also outperforms the narrow shell by 4.23 points; the shell-minus-harmonic interval is [-7.29,-1.11].

On the 10-composition confirmation panel, harmonic sources raise graph validity from 52.73% to 61.56%, a gain of 8.83 points (95% CI [5.08,12.50]; composition interval [4.84,12.66]). Geometric validity improves by 9.69 points [6.33,12.89]. Every graph-valid output has distinct perceived connectivity within its composition, and the evaluator encounters no exceptions. Generation takes nearly the same time for the two sources: harmonic/Gaussian timings are 175.73/175.43 seconds and 178.20/177.74 seconds for 640 samples in the two continuations.

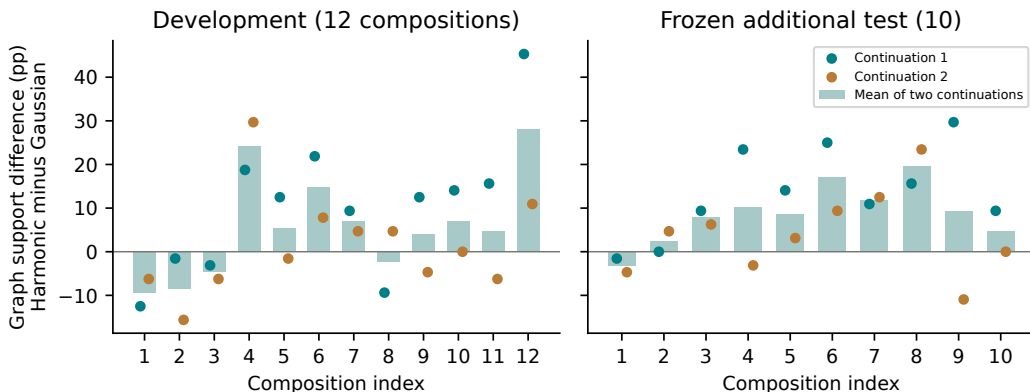


Figure 4: Harmonic-source graph-support gains vary across compositions and continuations. Points show individual continuation differences; bars average them. Both panels evaluate the same two trained models.

Table 6: Covariance-Gaussian control at matched continuation and inference budgets, using the calibrated 128-tree covariance estimate.

Panel	Continuation	Covariance Gaussian	Harmonic tree
Development	1	332/768	436/768
	2	313/768	388/768
Additional	1	286/640	413/640
	2	305/640	375/640

Control for source covariance. To test the role of second moments, we construct a Gaussian from a 128-tree estimate of the harmonic mixture covariance and symmetrize it over identical species. Calibration against an independent 8,192-tree estimate gives 1.85–6.25% relative Frobenius error across the 22 compositions. This control uses the same data, optimizer, network and sampling budgets. It is evaluated after the initial source comparison against the frozen harmonic checkpoints.

Harmonic sources improve graph validity by 11.65 points on development (paired 95% [8.59,14.78]; descriptive composition [8.07,15.43]) and 15.39 points on the additional panel (paired 95% [11.64,19.06]; composition [11.95,18.98]). All four point differences are positive. The covariance Gaussian is also worse than the isotropic Gaussian in both panels. The harmonic mixture thus retains an advantage over this calibrated second-moment control, consistent with a useful contribution from its mixture structure. The comparison adds 2,816 outputs and holds parameter count, initial self-conditioning weights, training rows and optimization fixed.

D VALIDITY BY MOLECULAR SIZE

The 64-composition panel has 16 compositions in each size stratum.

E ADDITIONAL GENERATOR EVALUATIONS

E.1 AN INDEPENDENTLY ADAPTED DIFFUSION BASELINE

We additionally adapt the official EGNN/EDM backbone (Hoogeboom et al., 2022) to the same composition-conditioned task. Positions follow its variance-preserving diffusion schedule, while clean atom categories are conditions during both training and sampling. The GEOM-pretrained checkpoint receives 30,000 updates on the same 3,000 OMol25 training structures used by each source continuation, with its original 10^{-4} learning rate. We evaluate both 128 network evaluations and the full 1,001-call sampler, including the final observation step, on all 24 fresh compositions.

Table 7: Raw chemical-graph validity on the 64-composition panel. Each size range contains 1,024 attempts per method across two continuations. Values are percentages; the final column reports percentage-point differences.

Atoms	Gaussian	Harmonic	Difference
17–28	49.12	55.08	+5.96
29–40	24.02	38.28	+14.26
41–52	16.50	21.29	+4.79
53–64	9.77	12.40	+2.64
All 64 compositions	24.85	31.76	+6.91

Table 8: Raw graph validity of the independently adapted conditional EDM, 768 attempts per continuation. The harmonic and force-update models in Table 2 reach 52.73% and 53.45%, respectively, at 128 calls.

Sampler	Continuation 1	Continuation 2	Pooled rate
EDM, 128 calls	103	100	13.22%
EDM, 1,001 calls	93	106	12.96%

This comparison tests adaptation to the same task and training data with a different pretrained architecture and objective. EDM has 2.38 million parameters and one denoiser pass per update; LatentSpring has 5.90 million and two. The longer EDM continuation takes 481.51/463.61 seconds, compared with approximately 570–590 seconds for the source-model continuations. Its 128-call sampler takes 48.54/47.94 seconds per 768 outputs, and its 1,001-call sampler takes 377.65/374.12 seconds. These timings distinguish call count from computational cost. Pretraining histories differ, and composition-overlap checks cover the processed OMol25 corpora. The matched Gaussian–harmonic experiments provide the controlled comparison of source choice.

F INDEPENDENT-BACKBONE COMPARISON

This study tests the source on an independent architecture with no pretraining. All methods use the official EDM equivariant graph neural network (EGNN) (Hoogeboom et al., 2022), with four message-passing layers, 256 hidden features and 2,381,566 parameters. Within each of two independently initialized repetitions, all methods start from identical weights and receive the same minibatches. Atomic types are fixed inputs throughout training and generation.

The common data comprise 20,000 OMol25 training rows spanning 8,323 compositions, plus 512 internal validation rows spanning 299 other compositions. Both sets use neutral organic singlets with 8–64 atoms. A composition hash separates training and internal validation. Reference selection requires the declared geometric and chemical-graph checks, without the additional scaffold-tree filter used for the pretrained-model study. None of the 64 evaluation compositions appears in either subset or either full processed corpus.

Each model receives 30,000 AdamW updates at learning rate 10^{-4} and batch size 32, totaling 960,000 training presentations. Batches contain equal-sized molecules; selecting atom-count buckets in proportion to their populations gives uniform training-row marginals. We clip the gradient norm at one. Generation uses an exponential moving average of network weights at the fixed final update, with decay $\min(0.999, (1 + s)/(10 + s))$ at optimizer step s . Internal validation at 5,000, 15,000 and 30,000 updates diagnoses learning and does not select a checkpoint.

The Gaussian and harmonic FM arms share the linear interpolation, atom matching and rotation pairing. Both predict velocity with a single EGNN evaluation and use 64 midpoint steps for 128 network evaluations; this comparison has no self-conditioning. The harmonic source uses the same probability law as the main model. For its factorized affinities, independently drawing the $N - 2$ entries of a Prüfer sequence with probabilities $u_i / \sum_j u_j$ samples exactly that tree law. Decoding the sequence gives the tree used to place atoms.

EDM predicts Gaussian coordinate noise under its original variance-preserving schedule. The position-conditional GAGA variant (Qu et al., 2026) truncates this schedule at the published GEOM setting of 650 out of 1,000 steps. Its initial Gaussian variance is $\alpha_t^2 v + \sigma_t^2$, where α_t and σ_t are the schedule’s signal and noise coefficients. The training-only estimate $v = 3.0232257866 \text{ \AA}^2$ averages squared centered coordinates per independent coordinate. The truncation setting is transferred from GEOM and is not optimized on OMol25 evaluation outcomes.

All models generate 32 raw structures per evaluation composition at 128 network evaluations. The complete EDM and GAGA schedules, including the final observation step, are also reported at 1,001 and 651 evaluations. No physical energy labels, bond supervision, energy-based selection or geometry relaxation enter this study. The A100, H200 and A100 MIG allocations are recorded in the scheduler metadata; matching training presentations and network-call counts does not match wall time.

Table 9: Independent diffusion models at both the common 128-evaluation budget and their full reverse schedules. Each initialization generates 2,048 outputs.

Method	Evaluations	Valid, seed 1	Valid, seed 2	Rate (%)
EDM	128	301	216	12.62
EDM	1,001	308	206	12.55
GAGA	128	321	249	13.92
GAGA	651	273	267	13.18

Energy and force evaluation of the frozen generators. We score every output on the ten-composition panel and each original reference with the same conservative eSEN checkpoint at its original charge and spin. We use the parity averages $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. Differences are paired within composition before averaging; energies per atom also subtract that composition’s reference energy, which cancels in the paired difference. There are 5,160 raw energy/force queries, including both inversions of the references in each continuation.

Force RMS means the square root of the per-atom mean squared force norm, then averaged over samples. The joint-validity statistic uses only paired outputs accepted by both methods, with 3–35 pairs per composition/continuation cell. All-attempt statistics also include graph-invalid outputs. We report both to separate the aggregate score from the score among jointly supported structures.

The aggregate improvement is concentrated in the first continuation. All-attempt energy differences are -0.08049 and -0.00070 eV/atom; force-RMS differences are -1.21996 and $+0.29302$ eV/Å, respectively.

Original endpoint-checkpoint reference. We evaluate the original 30,000-step OMol25 checkpoint on the coordinate-generation task by clamping composition and the neutral/no-bond inputs. Its endpoint sampler uses 127 Euler updates and a first-step bootstrap, totaling 128 network evaluations. Retaining predicted categorical history gives 106/640 and 113/640 graph-valid outputs in two sampling streams. Fixing the history categories and centering the first bootstrap gives 116/640 and 120/640. Both streams use the same checkpoint; the second variant changes both categorical history and bootstrap centering. The LatentSpring source models additionally receive shared task adaptation and 3,000 specialized updates. We therefore use the matched Gaussian and covariance-Gaussian continuations to assess the effect of source choice.

G NORMALIZATION OF THE HARMONIC MIXTURE

For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be its edge-incidence matrix, with one row per edge and entries $+1$ and -1 at the edge’s two endpoints. Let $U \in \mathbb{R}^{N \times (N-1)}$ have orthonormal columns spanning the centered subspace. Because $B_T \mathbf{1} = 0$, $(B_T U)(B_T U)^\top = B_T B_T^\top$. Cauchy–Binet expresses $\det(B_T B_T^\top)$ as the sum of squares of its N maximal minors. Each minor of a tree incidence matrix is ± 1 , so the determinant is N . The absolute Jacobian in one spatial dimension is therefore $\sqrt{N}$, and in three dimensions it is $N^{3/2}$. Consequently the component density is $q(X \mid T, c) =$

Table 10: Harmonic minus isotropic Gaussian physical readouts on the frozen additional panel. Brackets give paired 95% confidence intervals.

Readout	Support	Difference	Interval
Energy (eV/atom)	All attempts	−0.0406	[−0.0579, −0.0232]
	Both outputs valid	−0.0340	[−0.0692, −0.0003]
Force RMS (eV/Å)	All attempts	−0.4635	[−0.6782, −0.2512]
	Both outputs valid	−0.5389	[−0.8950, −0.2089]

$N^{3/2} \prod_{e \in T} h_e(B_T X)$ and integrates to one. Averaging with Equation (1) yields Equation (3) and preserves normalization.

H SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Suppose q_0 is G -invariant, $X \sim q_0$, and a pairing algorithm selects $g(X, Y) \in G$ using a target Y . If H is an independent Haar draw, then conditional on the orbit of X and on the selected g , HgX has the uniform orbit distribution of HX , by Haar invariance. Integrating over the unchanged source-orbit distribution recovers q_0 . This argument permits a non-Gaussian source and explains the shared augmentation used in the pairing. The paired endpoints remain correlated; our training objective uses their conditional displacement target directly.

I COVARIANCE CONTROLS

The shell control draws a positive edge length R concentrated around the radius-based scale ℓ , then chooses its direction uniformly. Specifically, $R = \ell \exp(sZ - s^2/2)$, where $Z \sim N(0, 1)$ is a scalar standard Gaussian and $s = 0.2$ controls the spread of log lengths. Its second moment is $\mathbb{E}R^2 = \ell^2 \exp(s^2)$. A uniform direction has second moment $I/3$, so each edge has covariance $vI = \ell^2 \exp(s^2)I/3$. Let A_T map edge coordinates to centered node coordinates. Independence gives $C_T = A_T \text{diag}(v_e) A_T^\top$. Here C_T describes correlations between atom positions for a fixed tree, and $\text{diag}(v_e)$ places its edge variances on a diagonal matrix. Replacing the edge vectors by independent Gaussians of covariance $v_e I$ leaves C_T unchanged and, since all component means are zero, leaves $\mathbb{E}_T C_T$ unchanged. The single-Gaussian control uses a finite, composition-seeded average of these matrices. Averaging this matrix over within-species permutations preserves positive definiteness on the centered subspace and yields species invariance. The Gaussian is exactly normalized for its chosen covariance; equality to the infinite tree-mixture covariance remains approximate.

J JARZYNSKI IDENTITY FOR ESCORTED TRANSPORT

Jarzynski’s relation connects exponential work averages with equilibrium free-energy differences, including in single-molecule pulling experiments (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport introduces an imposed flow and includes its configuration-volume correction in the work (Vaikuntanathan & Jarzynski, 2008). The change-of-variables form provides the basis for our local-reference construction.

Fix a composition and measure volume within its centered coordinate space. In this subsection, x denotes all independent coordinates of a molecule. Write $\beta = 1/(k_B T_{\text{phys}})$ for inverse thermal energy. The function $\rho_1(x) = \mathbf{1}_\Omega(x) \exp[-\beta E(x)]$ defines an unnormalized target density with finite, nonzero $Z_1 = \int_{\mathcal{H}_N} \rho_1(x) dx$, for example a declared confined molecular ensemble. The indicator $\mathbf{1}_\Omega$ equals one inside the allowed configuration region Ω and zero outside it. Restricting this region makes the integral finite. Dividing ρ_1 by Z_1 gives the target probability density. Let $q_0 > 0$ be the normalized source in Equation (3), and let $\Phi : \mathcal{H}_N \rightarrow \mathcal{H}_N$ be a differentiable bijection with nonzero intrinsic Jacobian. Here Φ is an imposed, invertible transformation of coordinates. Its Jacobian matrix $D\Phi$ contains the derivatives of output coordinates with respect to input coordinates; the absolute determinant measures local expansion or compression. For $\Phi(x_0) \in \Omega$, define the

generalized work

$$W_{\Phi}(x_0) = E(\Phi(x_0)) + \beta^{-1} \log q_0(x_0) - \beta^{-1} \log |\det D\Phi(x_0)|, \quad (7)$$

and set $W_{\Phi} = +\infty$ otherwise, assigning zero weight to an inadmissible configuration. Then

$$\mathbb{E}_{q_0}[e^{-\beta W_{\Phi}}] = \int_{\mathcal{H}_N} \rho_1(\Phi(x_0)) |\det D\Phi(x_0)| dx_0 \quad (8)$$

$$= Z_1. \quad (9)$$

Equivalently, the auxiliary source energy $E_0 = -\beta^{-1} \log q_0$ has $Z_0 = 1$ in the declared measure, yielding a Jarzynski-type normalizer identity. The Jacobian term accounts for compression by the escort map. For an observable f , such as energy or a function of a rotation angle, target averages are estimated by $\sum_j w_j f(\Phi(x_j)) / \sum_j w_j$, where $w_j = e^{-\beta W_{\Phi}(x_j)}$. This is self-normalized importance sampling: each transformed sample contributes according to its corrected weight. Finite-sample variance depends on how well the proposals cover the target distribution.

For an exact, sufficiently regular ordinary differential equation (ODE) flow, the intrinsic log determinant is the integral of the velocity divergence, which sums the local rates of expansion of independent coordinate directions. Our implementation makes the work tractable through explicit local references: translations of Gaussian anchor distributions in Appendix K, and periodic rotor maps with analytic Jacobians in Appendix L. The neural generator itself is evaluated through its raw samples, using the midpoint solver and terminal noise specified in Section 4.

K LOCAL FORCE-ESCORTED WORK DISTILLATION

The physical teacher combines local force-guided proposals with energy-weighted flow matching (Dern et al., 2025; Midgley et al., 2023). Normalized anchor references make the escorted-work weights explicit (Vaikuntanathan & Jarzynski, 2008).

Energy, force and mirror symmetry. The energy model assigns a scalar $E(X)$ to a geometry and provides its force $F(X) = -\nabla_X E(X)$, a vector for each atom. The force points in the direction of decreasing energy. We average a geometry and its inversion through the origin: $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. The opposite sign in the force average preserves $F_+ = -\nabla E_+$. This construction assigns equal energy to mirror-related geometries and opposite forces to their inverted coordinates. We remove the mean atom force before constructing displacements, keeping them in the centered coordinate space.

Local reference and target. For a generated training anchor A with accepted perceived graph G_A , define $q_A(X) = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. Write $kT = k_B T_{\text{phys}} = 0.0258519998$ eV for the thermal energy at 300 K. We set

$$\sigma_A = \min \left(0.03 \text{ \AA}, \sqrt{\frac{0.10 \text{ \AA} kT}{\|F_+(A)\|}} \right), \quad \delta_A = \frac{\sigma_A^2}{kT} F_+(A). \quad (10)$$

A vanishing force selects the maximal width. The force norm is the Frobenius norm over all atom-force components. These choices bound that same norm of the displacement by 0.10 Å and the Gaussian width by 0.03 Å. These quantities are fixed before drawing $X \sim q_A$. The escort $Y = X + \delta_A$ is a translation on the centered subspace and has determinant one. The local unnormalized target is

$$\rho_A(Y) = q_A(Y) \exp \left[-\frac{E_+(Y) - E_+(A)}{kT} \right] \mathbf{1}_{G_A}(Y), \quad (11)$$

where the support requires the same labelled perceived bond orders, formal charges and radical counts, together with the connected/nonoverlap assay. The graph is inferred from the anchor coordinates. The resulting target is a local energy tilt restrained around A by q_A .

Why the force displacement has this form. Locally approximate the molecular energy by its value and force at the anchor: $E_+(Y) - E_+(A) \simeq -\langle F_+(A), Y - A \rangle_F$, where the inner product sums

Table 11: Direct physics-distillation graph validity per 640 attempts. The frozen reference is not refitted; other arms have identical additional update budgets.

	Continuation	Frozen	Replay	Unweighted escort	Complete work
1		418	340	330	344
2		377	345	371	362

force times displacement over every atom and coordinate. Completing the square in the Gaussian gives

$$q_A(Y) \exp \left[\frac{\langle F_+(A), Y - A \rangle_F}{kT} \right] \propto \mathcal{N}_{\mathcal{H}_N} \left(Y; A + \frac{\sigma_A^2}{kT} F_+(A), \sigma_A^2 I \right). \quad (12)$$

Thus the force escort samples the local Boltzmann tilt exactly for a linearized energy, before applying the same-graph restriction. The Gaussian restraint acts like a virtual harmonic spring with stiffness kT/σ_A^2 : the force shifts its mean by force divided by stiffness. This explains the displacement scale. The graph restriction subsequently truncates this local distribution.

Complete local work. With reference Hamiltonian $H_0 = -kT \log q_A$ and target Hamiltonian $H_1 = H_0 + E_+ - E_+(A)$ plus hard support, the generalized quench work is

$$W_A(X) = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)] \quad (13)$$

on valid support, and $+\infty$ otherwise. Change of variables gives $\mathbb{E}_{q_A}[\exp(-W_A/kT)] = \int \rho_A(Y) dY$. The two reference-density terms correct for the escort’s proposal shift. Analytic checks recover constant work for a constant-force escort and the normalizer and weighted mean for a quadratic energy. For the linearized energy in Equation (12), substitution gives the same work $W_A = -\sigma_A^2 \|F_+(A)\|_F^2 / (2kT)$ for every supported candidate. Uniform weights are therefore appropriate in this approximation. Complete-work weights use the actual nonlinear energy to correct deviations from it. This distinguishes the inexpensive force-only teacher from its energy-weighted extension.

Training and controls. Each continuation selects eight distinct 17–28-atom training compositions by a fixed metadata hash and generates 32 anchors per composition. These compositions are disjoint from evaluation. Invalid anchors remain in the generation counts and are excluded from the teacher pool. Each valid anchor produces eight escorted particles. Particles outside its declared support receive zero weight. The work arm samples training endpoints using self-normalized $\exp(-W_A/kT)$ weights; the unweighted escort arm samples the same valid particles uniformly. A replay arm uses the retained raw anchors, and a frozen model supplies the reference. All-zero-particle anchors are recorded and excluded identically from all student pools.

Each student alternates 500 original-reference FM updates with 500 generated-anchor endpoint updates, sharing initialization, anchor draws, optimization and sampling budgets. Physical preparation costs differ: replay needs no force queries, the unweighted escort needs anchor forces, and complete work additionally needs energies at valid proposed endpoints. Actual shared research cost and these method requirements are reported separately. The final generator uses the learned network alone. Its temperature feature stays fixed throughout training and evaluation; 300 K determines the local teacher weights. Self-normalization introduces finite-particle bias, and the reported teacher ESS is computed over the eight local particles at each anchor. Effective sample size (ESS) is $(\sum_j w_j)^2 / \sum_j w_j^2$ for candidate weights w_j . It equals the candidate count for equal weights and approaches one when a single candidate dominates.

Direct-distillation outcome. Each continuation fits three students for 1,000 updates. Of 512 generated training anchors, 351 pass the graph assay and 349 have at least one supported escorted particle. Mean local particle ESS by composition is about 1.27–1.87 out of 8, indicating substantial weight concentration and limited finite-particle resolution of the local target.

Among paired outputs that are both graph-valid, complete work versus the unweighted escort reduces mean energy by 0.00758 eV/atom (conditional paired 95% interval $[-0.01424, -0.00113]$)

Table 12: Paired physical updates on the development panel. Energy differences are relative to the frozen model among jointly valid outputs, with paired 95% confidence intervals. This comparison follows the direct-distillation study.

Model	Graph-valid outputs (two streams)	Energy change (eV/atom)
Frozen	418/640; 377/640	0
Escort difference	407/640; 387/640	-0.01554 [-0.02084, -0.01031]
Work difference	406/640; 382/640	-0.01604 [-0.02188, -0.01013]

for the signed difference) and force RMS by 0.16174 eV/Å (interval [-0.27660, -0.05590]). Both continuations favor complete work. The all-output comparison has the same direction.

All three direct continuations reduce quality relative to the frozen model. Complete work has graph validity 55.16%, versus 62.11% for frozen, and jointly valid energy is *higher* by 0.10587 eV/atom (interval [0.09009, 0.12168]). Replay also regresses on the mixed training targets. This motivates transferring the physical update relative to its matched replay continuation, as described below.

Teacher preparation uses 6,232 raw energy/force queries: 702 at anchors and 5,530 at supported particles. An unweighted escort could omit the particle-energy queries; replay could omit all physical queries. Evaluating the four generated-output arms uses another 10,240 queries, reported separately from generation cost.

Transferring the paired physical update. We use parameter arithmetic (Ilharco et al., 2023) to transfer the difference between physical and replay continuations from the same initialization:

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}}), \quad (14)$$

for both unweighted-escort and complete-work students. The coefficient is fixed at one. Parameter tensors are updated, while buffers and the source distribution remain fixed.

Both physical-update variants lower energy and force in both continuations. The work-difference variant reduces force RMS by 0.38403 eV/Å among jointly valid outputs (interval [-0.48124, -0.28932]); all-attempt energy and force readouts also improve. Its observed graph-validity-rate difference is -0.55 percentage points, with interval [-2.58, 1.56].

The two physical-update variants have similar scores: their jointly valid energy difference is -0.00243 eV/atom, with interval [-0.00756, 0.00264], while all-attempt energy difference is +0.00022 with interval [-0.00557, 0.00649]. Force and graph intervals also span zero. The unweighted variant requires 702 anchor queries for teacher preparation, compared with 6,232 for complete work. Both retain the same 128-call neural inference procedure.

Evaluating the two transferred variants adds 2,560 generated outputs and 5,120 physical queries. The parameter transfer itself requires no optimizer steps. We use these development results to choose the force-only update for the new-composition evaluation in Section 6.4.

L MOLECULAR ROTOR WORK PROTOCOL

The three training molecules are C=C(N)N(C)C(N)=O, CCOC(C)(C)N, and CC(=O)NCC=O. Selection examines the first five rows of an existing training subset, accepts the first three distinct compositions with a methyl rotor preserving the labeled perceived graph on a 64-angle grid, and queries no energy during selection. Only methyl hydrogens rotate around the carbon-anchor axis; the remaining internal coordinates stay fixed. Overall translation is removed. The reference and target densities are defined with respect to the periodic angle measure $d\phi$ on $[-\pi, \pi)$.

The starting angular density is $q_0(\phi) = \exp(\cos \phi)/(2\pi I_0(1))$, where $2\pi I_0(1) = \int_{-\pi}^{\pi} \exp(\cos u) du$ is its normalizing integral. Define the reference Hamiltonian (energy function) $H_0 = -kT \log q_0$ and target Hamiltonian $H_1 = H_0 + U - U(0)$. The imposed map is $\psi_a(\phi) = \phi + a \sin \phi$, with $a \in \{-0.5, 0, 0.5\}$ controlling its strength. Its local volume factor is $J_a(\phi) = 1 + a \cos \phi$. The condition $|a| < 1$ holds, so it is a smooth, invertible transformation of the periodic angle. Its complete

work is

$$W_a(\phi) = U(\psi_a(\phi)) - U(0) + kT [\log q_0(\phi) - \log q_0(\psi_a(\phi)) - \log J_a(\phi)]. \quad (15)$$

Change of variables gives

$$\mathbb{E}_{q_0} e^{-W_a/kT} = \int_{-\pi}^{\pi} q_0(y) e^{-[U(y)-U(0)]/kT} dy = Z_1/Z_0, \quad Z_0 = 1. \quad (16)$$

The logarithmic Jacobian accounts for the compression or expansion caused by the imposed escort.

All 1,536 single-point calculations succeed. The reference potential is a periodic cubic interpolant of 512 GFN2 values per rotor. Its 256-knot interpolation error at the additional midpoints is at most 10^{-8} eV in this study. Increasing the reference quadrature from 32,768 to 65,536 points changes the free energy by less than 10^{-10} eV. These checks quantify interpolation and quadrature error for the GFN2-based reference potential.

Total variation uses 64 equal angular bins, with reference probabilities obtained by dense quadrature. Self-normalized estimates of target observables have finite-sample bias; the mean unnormalized weight is an unbiased normalizer estimator. The reported free energy is $-kT \log \hat{Z}$, with its finite-sample bias included in RMSE. At 2,048 draws, complete-work ESS ranges from 788 to 1,892 across the six nonidentity cases. Some incomplete weights have higher ESS despite their biased target. Correct target recovery therefore depends on the work definition as well as the concentration of the weights.

M PHYSICAL QUALITY OF THE MATCHED GENERATORS

We extend the independent EGNN comparison in Section 6.3 with physical readouts of its 16,384 saved 128-call outputs. The models, geometries, composition panel and generation budgets are unchanged. The readout protocol is fixed before any of these physical scores are evaluated. eSEN scores every output and its reflection; GFN2 scores every graph-valid output and both reference parities. All 2,198 GFN2 calculations succeed, including 256 reference checks, whose inversion tests pass. The eSEN readout uses 33,024 evaluations.

A quality yield is the fraction of *all attempted outputs* that pass the chemical-graph assay and a specified physical threshold. Invalid structures therefore remain in the denominator. Figure 5 varies the force-RMS threshold rather than assigning a single cutoff to physical validity. These thresholds describe the readout and do not certify a stationary geometry. The methods share the training budget and architecture specified in Appendix F; no physical update is added to these source and generator-objective controls.

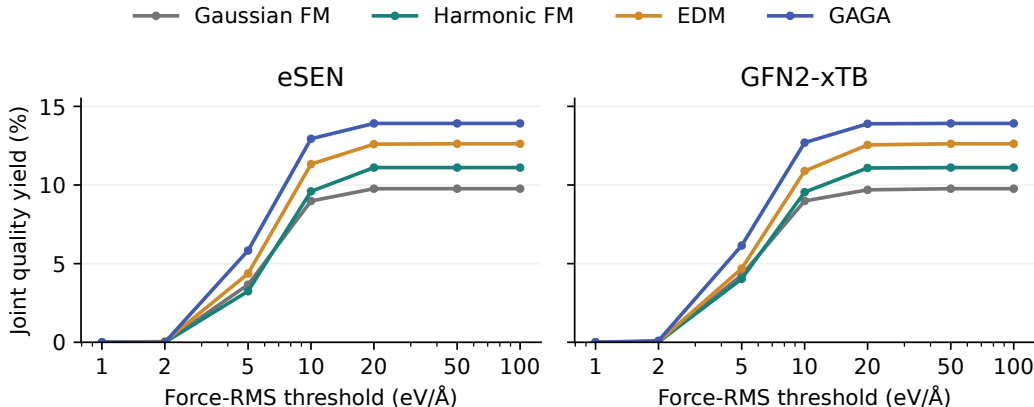


Figure 5: Fraction of all attempts that are graph-valid and have force RMS below the horizontal-axis threshold. Each curve pools the two existing initialization seeds and all 64 compositions. GAGA has the largest pooled yield at most nontrivial thresholds under both potentials.

Table 13: GFN2 readouts of the matched generators. Energy and force columns are medians over each method’s graph-valid outputs; yield uses all 4,096 attempts per method. Reference-relative energy subtracts the energy of one OMol25 geometry of the same composition. Generated connectivity may differ, so this quantity is not graph-specific strain energy.

Method	Energy excess (eV/atom)	Force RMS (eV/Å)	Yield at 5 eV/Å (%)
Gaussian FM	0.6820	5.2911	4.27
Harmonic FM	0.7114	5.8183	4.03
EDM	0.6799	5.8768	4.69
GAGA	0.6604	5.2785	6.15

Table 14: Self-conditioning ablation on the 24-composition panel. Each continuation has 768 raw attempts; the two counts give continuations 1 and 2. The full model improves over both single-pass budgets.

Model	Updates	Training passes	Valid outputs	Rate (%)
Single pass	3,000	3,000	363;336	45.51
Single pass	6,000	6,000	277;342	40.30
Self-conditioned	3,000	6,000	435;375	52.73

The source’s gain in graph validity does not imply lower physical scores among the resulting valid structures. GAGA also has a stronger aggregate physical readout in this matched setting. The physical-learning experiments separately test whether local physical targets improve a fixed harmonic generator.

N TRAINING-COMPONENT CONTROLS

N.1 WHAT SELF-CONDITIONING CONTRIBUTES

To test the provisional-geometry feedback, we train the harmonic-source model with self-conditioning removed. Initialization, the 3,000 training structures, source law and pairing are unchanged. One readout uses 3,000 optimizer updates; a second uses 6,000, matching the self-conditioned model’s number of backbone training evaluations. A single-pass model takes 64 midpoint steps, so all methods use 128 network evaluations at generation.

Self-conditioning adds 7.23 percentage points at equal optimizer updates (95% interval [4.69,9.77]) and 12.43 at equal training-pass count ([9.70,15.23]). Both continuations improve under both comparisons. Matching pass counts does not equate every operation or wall-clock cost; the comparison tests whether additional single-pass training recovers the observed validity gain.

N.2 WHY SUBTRACT THE REPLAY UPDATE

We compare the paired update with direct physical continuation and with a direct update scaled to the same parameter-space length. For this control, $\theta_{\text{control}} = \theta_{\text{base}} + \alpha(\theta_{\text{physical}} - \theta_{\text{base}})$, where α is the ratio of the paired and direct update norms. These norms sum squared changes over trainable parameters only. The resulting coefficients, 0.2536 and 0.2564, are determined from training weights before generation.

The same-norm control largely preserves graph validity but raises energy relative to the base model. The paired update instead lowers it. Directly comparing the paired and same-norm updates gives an energy change of -0.03232 eV/atom $([-0.03981, -0.02485])$ and a force-RMS change of -0.4170 eV/Å $([-0.5529, -0.2859])$. Both continuations favor the paired direction, and the all-output comparisons have the same signs. Their graph-validity difference is $+0.39$ points with interval $[-1.95, 2.73]$. The result supports the direction identified by replay subtraction, beyond simply limiting update magnitude.

Table 15: Physical-update controls on the ten-composition panel, using 1,280 attempts per model. Energy differences are relative to the base model under eSEN, among paired outputs that are both graph-valid.

Model	Graph validity (%)	Energy change (eV/atom)
Base	62.11	0
Direct physical update	54.77	+0.11080
Same-norm direct update	61.64	+0.01893
Paired update	62.03	-0.01554

Contribution of the force displacement. We also remove only the displacement from the local training targets, retaining the same Gaussian perturbations, candidate indices, widths, training schedule and paired parameter update. Compared with this control, the force-shifted targets lower energy by 0.01939 eV/atom (95% interval [-0.02417,-0.01455] for the signed difference) and force RMS by 0.3926 eV/Å ([-0.4696,-0.3263]) among jointly valid outputs. Both continuations improve. Graph validity is similar: the force-minus-control difference is +0.39 points with interval [-1.33,2.19]. The original force-dependent widths and candidate-selection rule remain fixed, so this comparison identifies the contribution of the displacement itself.

O CURVATURE-AWARE LOCAL WORK SAMPLING

The local force escort shifts every candidate in the same direction. Molecular energies also change more sharply along some displacement directions than others, such as bond stretching versus softer collective motion. We test whether estimating this local curvature improves the finite-particle work teacher while keeping its target distribution fixed.

Fitting the local energy shape. For this derivation, we flatten an anchor into a vector of $3N$ coordinates. The columns of $Q_N \in \mathbb{R}^{3N \times d}$ form an orthonormal basis of the centered coordinate space, where $d = 3(N - 1)$. A vector $u \in \mathbb{R}^d$ specifies an independent displacement in angstroms, and $Q_N u$ expresses it in the original atom coordinates. The vector f_A is the anchor force in the same basis. From eight independent pilot candidates, we collect displacement columns in D and corresponding negative force-change columns in G . A matrix B_A describes how force changes with displacement: $G \approx B_A D$. We fit a symmetric matrix by minimizing

$$\|BD - G\|_F^2 + \lambda \|B - \kappa I\|_F^2, \quad \kappa = \max\left(0, \frac{\langle D, G \rangle_F}{\|D\|_F^2}\right), \quad \lambda = 0.01 \|D\|_F^2 / K. \quad (17)$$

Here K is the number of graph-supported pilot columns, κ is their average positive stiffness, and the second term regularizes directions poorly resolved by the pilot. Projecting the fitted matrix onto the positive semidefinite cone replaces its negative eigenvalues by zero and defines B_A . The isotropic control instead uses $B_A = \kappa I$, giving every direction the same fitted stiffness. The pilot is held fixed before drawing production candidates.

An affine escort and its complete work. The local quadratic approximation is $E_+(A + Q_N u) - E_+(A) \approx -f_A^\top u + u^\top B_A u / 2$. Multiplying its Boltzmann factor by the original Gaussian reference gives a Gaussian with adjusted mean and covariance. Define the dimensionless precision matrix $P_A = I + \sigma_A^2 B_A / (k_B T_{\text{phys}})$. Its eigenvalues are positive. For a source displacement u , the new displacement is

$$v = P_A^{-1/2} u + b_A, \quad b_A = P_A^{-1} \frac{\sigma_A^2 f_A}{k_B T_{\text{phys}}}, \quad \log J_A = -\frac{1}{2} \log \det P_A. \quad (18)$$

The inverse square root reduces perturbations more strongly in stiff directions; the mean shift also accounts for this stiffness. The Jacobian J_A measures the map’s volume change in the d independent coordinates. We use the original reference width and temperature, so the restrained target is unchanged. With $X = A + Q_N u$ and $Y = A + Q_N v$, the complete work is

$$W_A = E_+(Y) - E_+(A) + k_B T_{\text{phys}} [\log q_A(X) - \log q_A(Y) - \log J_A]. \quad (19)$$

Candidates outside the original same-graph support receive zero weight. This is a local application of targeted and escorted work reweighting (Jarzynski, 2002; Vaikuntanathan & Jarzynski, 2008).

To see why the map helps, write the actual energy as the fitted quadratic plus a remainder $R_A(v)$. Substitution into the work formula gives $W_A = -k_B T_{\text{phys}} \log Z_{\text{quad}} + R_A(v)$, where Z_{quad} is the normalizing integral of the Gaussian reference times the quadratic Boltzmann factor. The fitted quadratic term cancels from the work; only the remaining energy mismatch varies across supported particles. If the energy equals the fitted quadratic, all supported particles have equal weights. Tests verify the constant-work identity for an anisotropic quadratic energy and verify rotation equivariance of the fitted escort.

Equal-budget molecular pilot. We select the first four eligible anchors from each of eight existing training compositions in each continuation, without energy or ESS ranking. The translation control uses 24 fresh production candidates per anchor. Each fitted escort uses at most eight pilot energy/force queries and 16 fresh production candidates. Thus its maximum particle-query budget equals that of the translation control; common anchor queries and reflection evaluations are accounted for separately. All production energies are evaluated, including graph-invalid candidates. Pilot samples are not reused as production samples for the fitted maps.

Table 16: Local effective sample size (ESS) under complete work at a maximum 24-particle query budget per anchor. Each continuation contains 32 anchors. The two fitted maps reserve eight queries for curvature estimation.

Escort	Production particles	ESS, continuation 1	ESS, continuation 2
Force translation	24	2.22	1.86
Isotropic curvature	16	3.55	3.94
Directional curvature	16	6.69	6.70

The directional fit improves ESS over translation on 61 of 64 anchors. Its mean gain is 4.66 effective samples, with a composition-bootstrap 95% interval of [3.81, 5.46]. Same-graph retention is 99.51%, versus 99.54% for translation. Independent multivariate-Gaussian density calculations reproduce the stored complete work weights. A subsequent full teacher construction uses all 349 original eligible anchors with fresh production draws; mean ESS is 2.32 versus 6.71 in the first continuation and 2.33 versus 6.37 in the second. These are local teacher diagnostics; the separate student comparison tests their effect on neural generation.

Transfer into the molecular flow. We train three physical students on these full teacher pools: uniform weighting of 24 force-translated candidates, complete-work weighting of the same 24 candidates, and complete-work weighting of 16 curvature-adjusted candidates. Every student follows the same 1,000-update schedule and reference/anchor choices as the original physical continuation. We reuse its matched replay model and transfer the physical-minus-replay update with coefficient one. All 349 original anchors remain eligible. The two work-weighted methods have the same maximum particle-query budget; uniform force targets can omit particle-energy queries. The main comparison uses the original preselected eight-particle force model; this follow-up separately tests the larger candidate pools and curvature fit.

Table 17: Raw student outputs on the 24-composition panel, with 1,536 attempts per method across the two continuations. Energy differences compare each work-weighted student with the uniform-force student on paired jointly-valid outputs. All methods use 128 network evaluations per output.

Teacher	Graph validity (%)	eSEN change (eV/atom)	GFN2 change (eV/atom)
Uniform force, 24 candidates	54.30	0	0
Complete work, 24 candidates	53.39	+0.00959	+0.00845
Curvature work, 16 candidates	53.32	+0.00793	+0.00783

Curvature-work versus translation-work energy changes are -0.00184 eV/atom under eSEN (95% interval $[-0.00654, 0.00289]$) and -0.00068 under GFN2 ($[-0.00484, 0.00341]$), with opposite

signs in the two continuations. The teacher’s ESS gain therefore does not establish an additional neural-generation benefit. Uniform force targets give lower energy and force than either work-weighted student under both potentials. Relative to the frozen harmonic generator, this uniform-force continuation lowers energy by 0.02088 eV/atom under eSEN and 0.01918 under GFN2. The primary generator retains the inexpensive force-based physical-learning recipe.